

Scalability & Future Plan

Date	7 July 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Supports only a limited number of simultaneous users.	Performance may decrease with high traffic.	High
2	Uses a local dataset and database.	Difficult to handle large-scale real-time data.	Medium
3	Prediction is based only on historical applicant data.	Does not consider real-time financial transactions.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports local users only.	Deploy on AWS/Azure with load balancing to support thousands of users.
2	Data Storage	Local CSV dataset.	Migrate to MySQL/PostgreSQL or cloud database for scalable storage.
3	Performance	Single Flask server with one ML model.	Optimize the model, implement caching, and deploy multiple server instances.
4	Security	Basic input validation.	Add user authentication, HTTPS, encryption, and secure API access using JWT/OAuth.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 2	Integrate real-time credit score and banking APIs.	2–4 Months	More accurate and up-to-date approval decisions.
Phase 3	Deploy the application on a cloud platform with scalable infrastructure.	4–6 Months	Better scalability, availability, and user experience.
Phase 4	Develop a mobile application and an admin dashboard with analytics and reporting features.	6–8 Months	Improved accessibility and efficient monitoring for financial institutions.